Session 22 · Tuesday 10 November 2026 · A home guide, worked after class. Item 1 is the controller we did not build and why it is not enough; item 2 is what the annihilation really is once a molecule has to do it; item 3 is their evidence and what the controller does not pin; item 4 is the ladder. Answers come on a separate sheet.

How to use this

Work it after class, on paper, before you touch PS8 Q4 and Q5. The answers are on a separate sheet — read them **after** you have written something down, not instead of writing something down.

These are the questions class did not have time for. Work them before you start PS8 — they set up the same machinery from a different angle.

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1. Why proportional feedback is not enough

Class derived that a controller obeying

$$\frac{du}{dt} = \mu - \theta X$$

drives the error to exactly zero, whatever plant it is attached to. The obvious simpler thing to try is **proportional** feedback — make the input depend on the error directly rather than on its accumulation:

$$u = \kappa\,(\mu/\theta - X)$$

Put this on session 5's plant, $\dot{X} = ku - \gamma X + d$.

(a) Solve for the steady state X_{ss} with $d = 0$.

(b) Now with $d \neq 0$. How much does the output move?

(c) You want the offset small. The algebra says to make κ large. Give two reasons — one from this course, one from the plant — that you cannot simply keep turning κ up.

(d) In one sentence: what does the integrator have that the proportional controller does not?

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2. What the annihilation has to be

The motif's third reaction is $Z_1 + Z_2 \to \varnothing$ at rate $\eta Z_1 Z_2$. Class showed the steady state does not contain η at all — not for large η, but for every $\eta > 0$.

(a) The derivation assumed Z_1 and Z_2 are lost **only** by annihilating each other. List every other way a protein is actually lost in a growing *E. coli* cell, and mark which of them remove Z_1 and Z_2 together, one for one, and which remove them independently.

(b) Aoki implement the annihilation with SigW and RsiW — a σ factor and its anti-σ factor — and the paper says RsiW "stably sequesters and holds SigW inactive." A sequestration complex is not literally destroyed; it

is bound. Write the reaction they actually have, with a complex C on the right.

(c) Under what condition on C does your reaction in (b) behave like the annihilation the derivation assumed? Name the condition, and name the session where you first met the thing that violates it.

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3. Reading their evidence

Aoki et al. 2019, Fig. 3d. They perturb temperature, 37 °C against 30 °C, and measure each strain's growth rate normalised to its own 37 °C rate, $n = 3$. Closed loop $P = 0.2261$; open loop $P = 7.769 \times 10^{-5}$.

(a) The closed-loop bars sit at 1.00 and 1.03; the open-loop ones at 1.00 and 0.33. Before any statistics: which of those two differences would you have believed from the picture alone, and what does the picture not tell you?

(b) The open-loop strain is the closed-loop circuit with the anti-σ module deleted. Say why that is a better control than (i) the same strain with no circuit at all, and (ii) a strain with the anti-σ module replaced by an inert protein of the same length.

(c) The controller holds the *mean* growth rate. Aoki report that the closed loop showed **increased cell-to-cell variability** compared with the open loop. Does that contradict the adaptation claim? Say what integral feedback pins and what it does not.

(d) They replaced sfGFP with *metE* in a Δ*metE* host, so the regulated output is the cell's growth rate. Why is that a harder test of the controller than regulating a fluorescent protein? Give the reason in terms of what the controller is now inside of.

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4. The ladder

Four sessions, one reaction.

(a) For each of sessions 18, 19, 21 and 22, name the molecule that was held and the thing that held it.

(b) On three of those four, the holding was a problem. On one it is the mechanism of the cure. Write the one sentence that is true of all four, and then say what changes sign in session 22.

(c) Part II's four devices each bought something and charged for it on a different axis. Fill the fourth row:

SESSION	WHAT YOU BOUGHT	WHAT IT COST
19	a budget you can hold to	measuring the burden, per construct
20	exact fluxes	an objective you had to import
21	a regulator that outruns its load	?
22	?	?